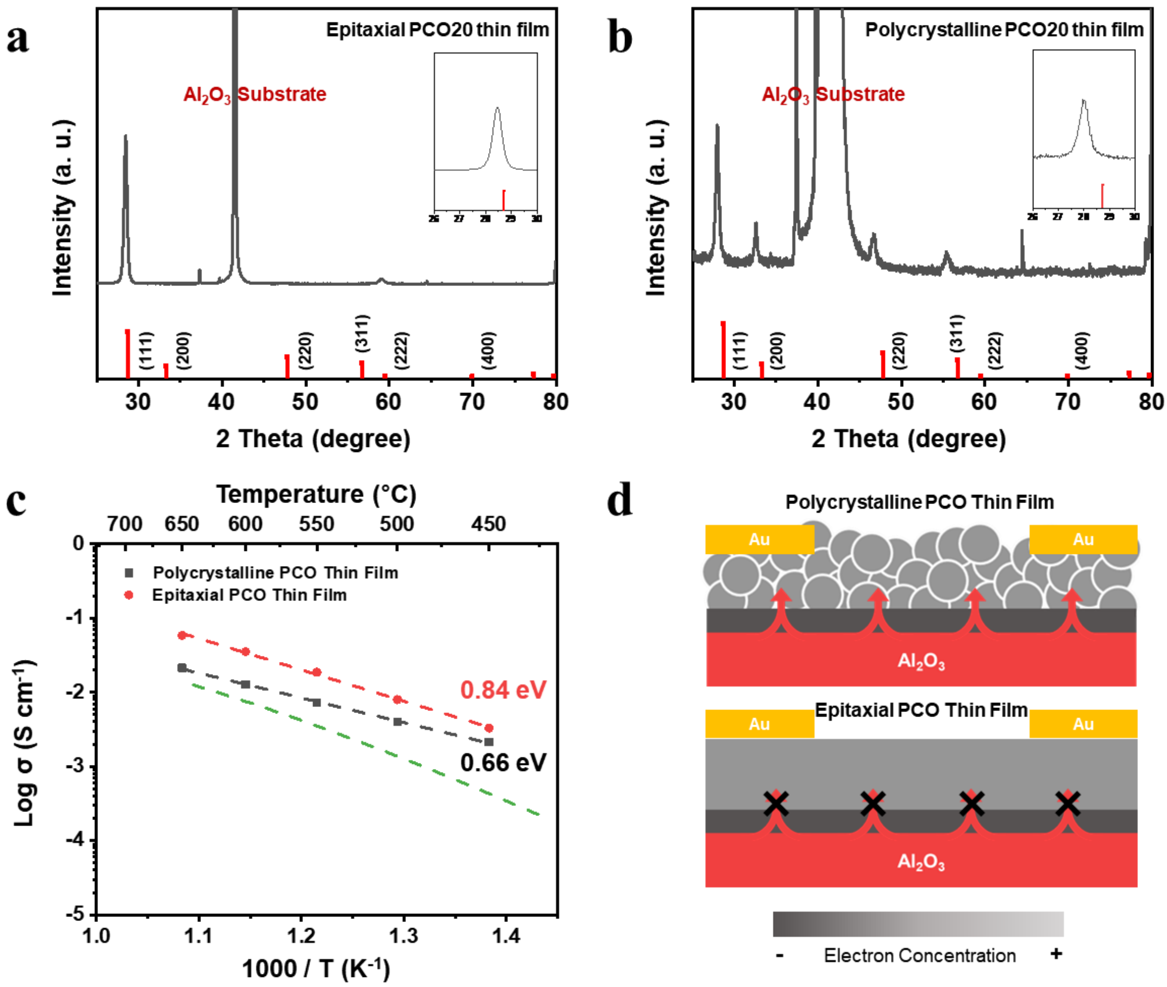

Supplementary Fig. 12 XRD spectra of **a** epitaxial and **b** polycrystalline PCO20 thin films deposited on Al_2O_3 insulating substrates via pulsed laser deposition (PLD). The inset images show enlarged views of the spectra from 26° to 30°, reavealing a peak shift due to tensile strain. Red vertical lines at bottom of figures indicate dominant facets of cerium oxide crystal structure. **c** Arrhenius plots of in-plane conductivity of epitaxial and polycrystalline PCO20 thin films, along with their respective activation energies. The green dashed line indicates the conductivity behavior of a bulk PCO20 film calculated with the aid of defect chemical modelling. **d** Comparison between polycrystalline and epitaxial PCO20 thin films. Substrate heterointerface shown in black corresponds to electron depletion due to relative acidity of Al_2O_3. Diffusion of Al ions only occurs in the polycrystalline PCO20 thin film, leading to GB electron accumulation, depicted in white.

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